



MADHU MARAJU

Chodavaram, Anakapalli, India

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EDUCATION

Bachelor of Technology in Mechanical Engineering	2024 - 2027
Vishnu Institute of Technology (A), Bhimavaram CGPA: 9.09 (Gold Medalist)	
Diploma in Mechanical Engineering	2021 - 2024
Government polytechnic, Tadepalligudem GPA: 9.4 (Gold Medalist)	
Secondary School Education	2021
Government High School, Chodavaram GPA: 10	

INTERNSHIPS

INDIAN INSTITUTE OF TECHNOLOGY PALAKKAD (IIT PKD)	<i>MAY-JUN 2026</i>
<i>SUMMER RESEARCH INTERN (Stipend ₹12000/-) – WELDING TECHNOLOGY LAB</i>	Kerala, India
- Designed TPMS-based heat exchanger, regenerative rocket nozzle, and conformal cooling mold using Fusion 360 and nTop following DfAM principles. - Implemented lattice structures and conformal cooling channels, and fabricated heat exchanger and rocket nozzle demonstration models through polymer-based 3D printing.	
INDIAN INSTITUTE OF TECHNOLOGY BHUBANESWAR (IIT BBS)	<i>DEC 2025</i>
<i>WINTER RESEARCH INTERN – SCHOOL OF MECHANICAL SCIENCE (saNDY Group)</i>	Bhubaneswar, India
Developed a mathematical model of a vibro-impact system incorporating frictional effects for vibration control applications. Implemented non-dimensional governing equations in MATLAB and conducted numerical simulations. Analyzed system dynamics using time-history responses, phase portraits, and bifurcation diagrams.	
INDIA YAMAHA MOTORS PVT.LTD.	<i>JUNE-JULY 2025</i>
<i>SUMMER INTERN – MANUFACTURING ENGINEERING (EXPORT)</i>	Chennai, India
Contributed to CKD IPC digitalization and production process optimization, achieving ~10% cycle time reduction. And gained hands-on exposure to casting, machining, welding, painting, assembly, and basic CNC programming while supporting production operations.	
SCHWING STETTER INDIA PVT.LTD.	<i>FEB-MAR 2025</i>
<i>ACADEMIC INTERN – PRODUCTION ENGINEERING</i>	Chennai, India
Analyzed concrete batching plant assembly workflows and identified process improvements with ~12% efficiency gain potential. Studied the working principles of load cells and proximity sensors, and gained practical exposure to industrial safety practices including 5S methodology.	
THE CHODAVARAM CO-OPERATIVE SUGARS LTD	<i>NOV 2023-MAY 2024</i>
<i>DIPLOMA TRAINEE – WORKSHOP</i>	Anakapalli, India
Assisted in sugar and co-generation plant operations and equipment maintenance, contributing to improved reliability of crushers, turbines, and boilers with reduced operational downtime.	

PROJECTS

ASSET INTEGRITY FRAMEWORK FOR LIFE EXTENSION OF AGEING INDUSTRIAL ASSETS	<i>(Team Lead of 5)</i>
Developed a predictive maintenance framework using multi-parameter sensing, Health Score logic, and real-time monitoring dashboards, supported by ANSYS creep and fatigue analysis for condition-based inspection; secured Runner-Up (₹1,00,000) at the ASME IMECE India 2025 Brain Bolt Event.	

HEAVY LIFT SYSTEM DESIGN AND STRUCTURAL ANALYSIS | ASME India 2026 – Semi Finalist

- Designed a 45-tonne low-headroom lifting system using a fixed spreader bar and twin cantilever beams.
- Performed structural verification including CoG, sling, beam, and ballast calculations as per ASME lifting standards. And validated the design through CAD, ANSYS FEA, and lattice optimization.

PATIENT- SPECIFIC PROSTHETIC SOCKET USING 3D PRINTING

(Team Lead of 6)

- Selected among the **Top 15 out of 305 teams** in **SIH 2025** Internal Hackathon for developing a patient-specific prosthetic socket using Fusion 360, ANSYS, and FDM 3D printing based on MRI data.
- Fabricated a PLA based socket, reducing fabrication time by 80% and cost by 70% while enhancing comfort and strength.

SMART GLASS CLEANING ROBOT | IMECE India 2025

(Team Lead of 5)

- Designed a remote-controlled glass cleaning robot for high-rise buildings within 100 cm³ size constraints, integrating suction-based climbing, spray-roller cleaning and IoT-enabled wireless control and designed a lightweight carbon-fiber chassis using generative design.

ELECTROCHEMICAL ENERGY HARVESTING FROM SOLAR STILL

(Team Lead of 5)

- Designed and developed a hybrid solar still combining water purification and electricity generation. Enhanced solar still performance by integrating electrochemical cells, delivering 0.8–1.2 V per cycle and ~25% improvement in water output.

SKILLS

- **CAD Tools:** AutoCAD | CATIA V5 | Fusion360 | SolidWorks | Rhino3D
- **Simulation & Analysis Tools:** ANSYS (structural analysis and CFD) | MATLAB | nTop
- **Presentation tools:** MS Word | MS Excel | MS PowerPoint | Canva.
- **Soft Skills:** Technical Documentation | Adaptability | Team Collaboration
- **Languages:** English | Telugu (Native) | Tamil | Japanese (Beginner level) | Hindi

CERTIFICATIONS

- Manufacturing Technology I & II (Elite) – [NPTEL](#), Operations Management (Elite) – [NPTEL](#)
- **MATLAB** onramp completion certificate – [MATLAB](#).
- **3D Printing** Fundamentals – [HP LIFE](#).
- [CFD & Heat Transfer](#) 3 - Day Workshop – BVRIT Narsapur, Conducted by IIT JAMMU Experts.

ACHIEVEMENTS

- **Runner-Up (2nd Prize), ASME IMECE India 2025** – Brain Bolt Event, Hyderabad, India | 10th – 12th September 2025.
- **Finalist** – Space Technologies for Earth & Beyond 2025: Viksit Bharat Ideathon ([IIT Tirupati](#)) – jointly organized by **ISRO–NRSC**, Tirupati, India.
- **Top 10 Finalist**, Regional-Level Techfest 2022, Vijayawada, India | November 2022 – Qualified for the [National-Level Competition](#).
- Secured Second Prize (₹2,000), **Researcher's Day 2026** – Poster Presentation on Nonlinear Vibration Control (VINES), Vishnu Institute of Technology, Bhimavaram, India | 2026.
- Secured Second Prize (₹2,000), **InnovIT-2K26 Project Expo**, Vishnu Institute of Technology, Bhimavaram, India | 2026.